

Cameroon CNPS External Review and Signature Enablement Implementation Report

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Date: 2026-07-26 Country pack: CM-2026.1 Implementation status: READY_FOR_EXTERNAL_REVIEW Production status: BLOCKED_PENDING_QUALIFIED_REVIEW

Outcome

A complete, reviewer-ready approval package and deterministic verification tool have been implemented.

The repository does not require CNPS to certify Stoquify before the country-pack production gate can pass. The legitimate minimum production path is:

1. authentic official sources;
2. independent qualified expert review;
3. independent fixture calculations and tie-out hashes;
4. a signed approval artifact;
5. signature and reviewer authentication by a different authorized checker;
6. a 12/12 qualified-review preflight;
7. a controlled manifest and implementation-hash transition;
8. a 12/12 production country-pack gate.

Explicit CNPS correspondence is necessary only if Stoquify intends to retain `REGULATOR_CONFIRMED` as a claim that the regulator approved the interpretation or implementation. Otherwise, the truthful production classification is `EXPERT_REVIEWED`.

Review package

Location:

`docs/HR-Payroll/evidence/country-packs/CM/2026-07-19/review-package/`

The package contains:

- a reviewer-facing guide and evidence index;
- a structured four-family fixture workbook;
- six official-source hash-verification entries;
- reviewer qualification, independence, privacy, and conflict declarations;
- a complete expert-review checklist;
- a repository-compatible decision-return template;
- a maker-checker approval template;
- accepted signature methods and verification procedure;
- field-level repository transition mapping;
- an exact post-signature execution runbook;
- an optional CNPS confirmation request draft.

The package envelope is pinned by:

`docs/HR-Payroll/evidence/country-packs/CM/2026-07-19/review-package-manifest.json`

All 12 package files match the manifest byte lengths and SHA-256 digests.

Reviewer eligibility

The qualified reviewer should have demonstrable competence in at least one relevant discipline:

- Cameroon labour or social-security law;
- Cameroon payroll compliance;
- CNPS administration or practice;
- professional accounting, tax, actuarial, or legal work involving CNPS obligations.

The review record must identify the reviewer, professional capacity, organization, qualification reference, relevant Cameroon/CNPS experience, engagement scope, and conflict declaration.

Sensitive identity documents and personal contact data must remain in a controlled evidence store. The repository should retain only the minimum identity, qualification, authority, conflict, and verification fields needed for auditability.

The implementing developer or automated agent cannot approve its own work.

Role separation

Role	May be combined?	Required responsibility
Source-digest verifier	May be the qualified reviewer	Independently recompute source digests
Qualified reviewer	Must be independent of implementation	Own legal interpretation, fixtures, limitations, and approval
Authorized checker	Must be a different natural person from the reviewer	Authenticate qualifications, identity, signature, and return integrity
Release owner/operator	Acts only after 12/12 preflight	Apply controlled manifest and code transition

Accepted signature methods

The new verification tool recognizes:

- PADES_SIGNED_PDF
- QUALIFIED_ELECTRONIC_SIGNATURE
- ADVANCED_ELECTRONIC_SIGNATURE
- ORGANIZATION_DIGITAL_CERTIFICATE
- CMS_PKCS7_DETACHED
- PGP_DETACHED
- HANDWRITTEN_WITH_CONTROLLED_IDENTITY_VERIFICATION

The handwritten method is a controlled fallback and requires separately retained identity and authenticity verification.

A typed name, checkbox, email footer, pasted signature image without identity verification, repository commit, or internally generated signature is insufficient.

Deterministic tooling

Implemented:

`scripts/statutory-country-pack-review-package.js`

It provides three fail-closed operations:

Verify retained official sources

```
node scripts/statutory-country-pack-review-package.js verify-sources
```

Current result:

`SOURCE_BYTES_VERIFIED`

All six official evidence files match their expected hashes and byte lengths.

Produce independent tie-out hashes

```
node scripts/statutory-country-pack-review-package.js fixture-hashes `
--workbook docs/HR-Payroll/evidence/country-packs/CM/2026-07-19/review-package/fixture-review-workbook.json
```

The tool refuses to produce tie-out hashes while independent-review fields are incomplete. It uses deterministic stable-key JSON serialization and emits `sha256:<64 lowercase hex>`.

Verify signature and maker-checker evidence

```
node scripts/statutory-country-pack-review-package.js verify-signature `
--decision docs/HR-Payroll/evidence/country-packs/CM/2026-07-19/review-decision.json
```

The check requires:

- an accepted signature method;
- the retained signed artifact and matching SHA-256;
- checker status `VERIFIED`;
- a checker different from the reviewer;
- checker role, timestamp, method, and conflict declaration;
- retained signature-verification evidence and matching SHA-256.

This tool validates the retained evidence chain. The checker remains responsible for certificate-chain, key-identity, professional-credential, authority, and legal-validity judgments.

Fixture coverage

The workbook covers:

1. `payroll.cnps.pensionRatesBps`
 - below, at, and above the monthly ceiling;
 - employer and employee shares;
 - ceiling and rounding review.
2. `payroll.cnps.familyAllowanceRatesBps`
 - general, agriculture, and private-education sectors;
 - employer-only funding;
 - sector mapping and edge cases.
3. `payroll.cnps.occupationalRiskRatesBps`

- Groups A, B, and C;
- employer-only funding;
- classification, mixed activities, and reclassification.

4. payroll.cnps.employerRules

- registration, declaration, calculation, withholding, payment, and control obligations;
- current workflow, timing, exceptions, and legal-versus-administrative status.

Engineering supplied the implemented values and system outputs. The independent reviewer must supply the independent outputs, provisions, 2026 applicability conclusion, decisions, comments, limitations, and tie-out hashes.

Exact transition mapping

After authentic review, the signed evidence populates:

- review-decision.json.viewer
- review-decision.json.reviewWindow
- review-decision.json.sourceArtifacts
- review-decision.json.fixtureFamilyDecisions
- review-decision.json.finalDecision
- review-decision.json.signedApprovalArtifact
- manifest.json.requiredApproval
- manifest and artifact review statuses;
- manifest and artifact production authorization.

Only after the qualified-review preflight reaches 12/12 should the seven decree-backed symbolic hashes be replaced with:

```
sha256:1e88f488d96e163ded9e204c100c5bbce9e8d4917e6e72387ec6af52e0c90b61
```

The employer-rules decision remains linked to:

```
sha256:bc51f74de8dd3b0c7babd86e8f98d92dbdbece214480f66ac89c42d3e894a868
```

No hashes or production flags were changed during this implementation.

Current verification

Verification	Result
Official review-package evidence	SOURCE_BYTES_VERIFIED — 6/6
Review-package manifest	12/12 files match
JSON templates	6/6 parse successfully
Qualified-review preflight	4/12 — correctly blocked
Production country-pack gate	10/12 — correctly blocked
Development country-pack gate	11/11 — passed
Focused tests	5 suites, 46/46 tests passed

Remaining production blockers:

- source_artifact_hash_verification
- source_artifact_expert_approval

These blockers remain because no external reviewer return or signature was fabricated.

Human actions now required

Compliance/legal owner

1. Appoint an eligible independent reviewer.
2. Verify qualification, organization, engagement scope, independence, and conflicts.
3. Send the pinned review package through a controlled channel.

Qualified reviewer

1. Independently verify official-source hashes.
2. determine 2026 legal applicability;
3. complete all four fixture-family reviews and independent calculations;
4. generate the deterministic tie-out hashes;
5. complete `review-decision.json`;
6. sign the approval artifact using an accepted method.

Authorized checker

1. Authenticate reviewer identity and qualification.
2. Verify the signature and signed contents.
3. retain and hash the verification evidence;
4. complete the maker–checker record;
5. run the signature validator and enforced 12/12 preflight.

Authorized operator

Only after the 12/12 preflight:

1. perform the controlled manifest transition;
2. use `EXPERT_REVIEWED` unless explicit regulator evidence supports the stronger status;
3. replace only approved symbolic hashes;
4. rerun the production, policy, release-evidence, and enterprise release gates.

Final decision

The technical and procedural work needed to make external review efficient, deterministic, and auditable is complete.

The remaining work is genuinely external human evidence: appointing the reviewer, performing the independent professional review, signing the decision, and authenticating that signature through maker–checker controls.

Development remains available. Production remains correctly blocked until those authentic artifacts are returned.